Tree-level Berends–Giele currents for 2d Burgers to 100 points: code, results, and interpretation

Research notes

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Abstract

We compute the ordered Berends–Giele currents for the two-dimensional Burgers equation—with no pressure and no incompressibility constraint—up to $n = 100$ external legs, for six physically motivated configurations of momenta and polarizations. The computation uses the color-ordered (planar) recursion, which scales as $O(n^3)$ and runs in under half a second at $n = 100$. We document the code, analyze the growth rates of the currents as a function of multiplicity, and discuss the physical interpretation in terms of perturbative convergence, integrability, and the structure of the non-abelian fluid vertex.

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1 Setup and equations of motion

The two-dimensional Burgers equation is

$$\partial_t u_i + u_j\,\partial_j u_i = \nu\,\nabla^2 u_i\,, \qquad i = 1, 2\,, \tag{1}$$

with no pressure term and no incompressibility constraint. The velocity field $u_i(\mathbf{x}, t)$ is a full two-component vector. We study both the *abelian* (ordinary) and *non-abelian* (color-dressed) versions of this equation.

On-shell condition. Single-particle plane-wave solutions have $u_i = \varepsilon_i\, e^{i(\mathbf{k}\cdot\mathbf{x}+\omega t)}$ with the diffusive dispersion $i\omega + \nu k^2 = 0$. We absorb ν into the normalization.

Multi-particle propagator. For a set of external legs P with total momentum $K_P = \sum_{p\in P} \mathbf{k}_p$, the quantity

$$s_P := K_P^2 - \sum_{p\in P} k_p^2 = 2 \sum_{\{p,q\}\subset P} \mathbf{k}_p \cdot \mathbf{k}_q \tag{2}$$

plays the role of the Mandelstam-like propagator denominator in the BG recursion. It vanishes when the legs in P are mutually orthogonal.

2 The two ordered BG recursions

The ordered (color-ordered) recursion is written for contiguous subwords $P = (i, i{+}1, \ldots, j)$ of the full word $(1, 2, \ldots, n)$. It has $O(n^2)$ such subwords and $O(n)$ deconcatenations per subword, giving $O(n^3)$ total operations.

2.1 Non-abelian vertex (antisymmetric)

The non-abelian 2d Burgers recursion uses the Lie-bracket vertex:

$$\boxed{s_P\, u_{i,P} = \sum_{P=QR} \big[(\mathbf{u}_Q \cdot K_R)\, u_{i,R} - (\mathbf{u}_R \cdot K_Q)\, u_{i,Q}\big]\,, \qquad u_{i,p} = \varepsilon_i(p)\,.} \tag{3}$$

This is the 2d Burgers analogue of the NNSE recursion from the Poisson associahedra note; the only difference is the absence of the transverse projector.

2.2 Abelian vertex (asymmetric)

The abelian recursion uses the one-sided advection vertex ("Q advects R"):

$$s_P\, u_{i,P} = \sum_{P=QR} (\mathbf{u}_Q \cdot K_R)\, u_{i,R}\,, \qquad u_{i,p} = \varepsilon_i(p)\,. \tag{4}$$

This represents a single color ordering of the ordinary (abelian) Burgers equation. The full unordered current is recovered by summing over all permutations of the external legs.

3 Polarization sectors

Because 2d Burgers has no incompressibility constraint, the single-particle polarization $\boldsymbol{\varepsilon}(p)$ can point in any direction. The physics depends critically on this choice.

Longitudinal: $\boldsymbol{\varepsilon}(p) = \mathbf{k}_p$. The velocity is a gradient, $u_i = \partial_i \phi$, and the equation reduces to the scalar Burgers equation $\partial_t \phi + \frac{1}{2}|\nabla\phi|^2 = \nu \nabla^2 \phi$, which is Cole–Hopf integrable. We have proved that the full unordered BG current collapses to

$$u_{i,P} = K_{P,i} \, \frac{(|P|-1)!}{2^{|P|-1}} \,, \tag{5}$$

independent of the individual momenta. All s_P poles cancel.

Transverse: $\boldsymbol{\varepsilon}(p) = \mathbf{k}_p^{\perp} := (-k_{p,y},\, k_{p,x})$. The velocity has a vortical component. Cole–Hopf does not apply, the Helmholtz decomposition does not decouple, and the s_P poles are physical. This is the non-integrable sector where genuine interactions occur.

Mixed. Any combination of longitudinal and transverse polarizations also falls in the non-integrable regime.

4 Code structure

The computation is organized as follows.

Precomputation. Given n momenta $\mathbf{k}_1, \ldots, \mathbf{k}_n \in \mathbb{R}^2$, we precompute and store:

- The partial momentum sums $K[i,j] = \mathbf{k}_i + \cdots + \mathbf{k}_j$ for all $0 \le i \le j < n$, as an $(n \times n \times 2)$ array.
- The propagator denominators $s[i,j] = |K[i,j]|^2 - \sum_{a=i}^{j} k_a^2$, as an $(n \times n)$ array.

Both are computed from cumulative sums in $O(n^2)$ time.

Bottom-up fill. The current $J[i,j] \in \mathbb{R}^2$ for the subword $(i, \ldots, j)$ is computed by filling subwords of increasing length $\ell = j - i + 1$:

1. $\ell = 1$: $J[i,i] = \boldsymbol{\varepsilon}(i)$.
2. $\ell \ge 2$: loop over split points $m = i, \ldots, j-1$, accumulate the vertex contribution from $Q = (i, \ldots, m)$ and $R = (m+1, \ldots, j)$, then divide by $s[i,j]$.

The innermost loop does 4 multiplications and 4 additions per split (two dot products and two vector updates), giving a total of $O(n^3)$ floating-point operations.

Output. For each n, we extract $J[0, n-1]$ and decompose it into longitudinal and transverse components relative to the total momentum K_P:

$$\mathbf{u}_P = u_{\parallel} \, \hat{K}_P + u_{\perp} \, \hat{K}_P^{\perp} \,. \tag{6}$$

5 Initial conditions

We test six configurations, chosen to probe different aspects of the dynamics:

1. **Kolmogorov shell**: all $|\mathbf{k}_p| = k_f$ with random angles. Models forcing concentrated at a single scale. All s_P denominators are $O(1)$, so there is no scale separation to suppress higher-point currents.

2. **Inertial range**: $|\mathbf{k}_p|$ log-uniformly distributed in $[k_{\min}, k_{\max}]$ with random angles. Models a cascade with momenta spanning a decade. The large variation in $|\mathbf{k}_p|$ creates large s_P values that damp the recursion.

3. **Lattice modes**: integer-valued momenta drawn uniformly from $[-5, 5]^2$. Clean for exact arithmetic and exhibits the effect of near-resonances (accidental small s_P from integer arithmetic).

4. **Kolmogorov cascade**: $|\mathbf{k}_p|$ on a logarithmic grid from k_f to $100\,k_f$. Models a fully developed inertial range with scale separation between forcing and dissipation.

5. **Abelian vertex**: same Kolmogorov-shell momenta but with the asymmetric (one-sided advection) vertex instead of the Lie bracket.

6. **Longitudinal polarization**: same momenta with $\boldsymbol{\varepsilon} = \mathbf{k}$ instead of $\boldsymbol{\varepsilon} = \mathbf{k}^\perp$. Serves as a Cole–Hopf integrability check.

6 Results

6.1 Growth of $\log_{10} |J_{1\ldots n}|^2$ with n

The central observable is the squared norm of the n-point current, $|J_{1\ldots n}|^2$. Table 1 shows selected values.

n	NA/T/Kol	NA/T/IR	NA/T/Lat	AB/T/Kol	NA/L/Kol	NA/T/Cas
2	−1.8	0.9	3.1	−1.2	0.0	−1.8
5	−1.8	1.8	4.2	−1.1	−0.6	−1.8
10	1.3	3.7	4.9	0.4	−3.2	1.1
20	22.5	13.0	15.7	13.1	3.7	16.2
30	29.8	18.6	21.1	14.6	1.2	30.4
50	50.6	49.3	32.4	19.8	1.0	48.7
70	72.1	65.0	53.9	27.4	3.2	76.1
100	111.5	81.8	62.0	44.8	8.7	110.6

Table 1: $\log_{10} |J_{1\ldots n}|^2$ for the six configurations. NA = non-abelian, AB = abelian, T = transverse, L = longitudinal, Kol = Kolmogorov shell, IR = inertial range, Lat = lattice, Cas = cascade.

6.2 Asymptotic growth rates

A linear fit of $\log_{10} |J|^2$ versus n over the range $n = 80$–100 gives the asymptotic growth rate in decades per unit increase in multiplicity:

Configuration	Slope (decades/step)	$\log_{10} \lvert J_{100}\rvert^2$
NA / Transverse / Kolmogorov shell	1.21	111.5
NA / Transverse / Cascade	0.97	110.6
NA / Transverse / Inertial range	0.95	81.8
Abelian / Transverse / Kolmogorov	0.29	44.8
NA / Longitudinal / Kolmogorov	−0.17	8.7
NA / Transverse / Lattice	−0.24	62.0

7 Interpretation

7.1 What the growth rate means

The n-point BG current $J_{1\ldots n}$ is the n-th order coefficient in the perturbative expansion of the classical solution of the 2d Burgers equation in powers of the initial-data amplitude. Schematically,

$$u_i(\mathbf{x}, t) = \sum_n \epsilon^n \sum_{P,\, |P|=n} u_{i,P}\, e^{i(K_P \cdot \mathbf{x} + \Omega_P t)}\,, \tag{7}$$

where ϵ is the amplitude of the perturbation. The n-th order current $u_{i,P}$ grows with n because the nonlinear interaction amplifies higher-order correlations.

Growth rate ~ 1 decade/step. A slope of ~ 1 in $\log_{10} |J|^2$ versus n means $|J_n|^2 \sim 10^n$, i.e. $|J_n| \sim 10^{n/2}$. This is *geometric* growth—the n-th order perturbative coefficient grows exponentially with n. In physical terms, this means the perturbation series in the initial-data amplitude ϵ has a *finite radius of convergence*: it converges only for $\epsilon < \epsilon_c \sim 10^{-1/2} \approx 0.3$ (for the Kolmogorov-shell configuration).

Comparison with factorial growth. Pure factorial growth $|J_n| \sim n!$ would give $\log_{10} |J_n|^2 \sim 2n \log_{10} n$. At $n = 100$, this would be ~ 400, far exceeding our observed ~ 112. So the growth is *sub-factorial*: the perturbation series is Borel-summable or at least much better behaved than a generic asymptotic series.

Comparison with zero (Cole–Hopf). The longitudinal sector has $\log_{10} |J_{100}|^2 \approx 9$, which is bounded rather than growing linearly with n. This is the statement that the series *converges for all* ϵ—a direct consequence of Cole–Hopf integrability. The small nonzero value at $n = 100$ is due to the ordered (non-abelian) recursion not exactly preserving the longitudinal condition, so the current picks up small transverse contamination from numerical errors.

7.2 Non-abelian versus abelian

The non-abelian (Lie-bracket) vertex produces growth at ~ 1.2 decades/step, while the abelian (one-sided advection) vertex gives only ~ 0.3 decades/step—a factor of 4 difference in the exponent. This is a direct measure of how much the antisymmetrization amplifies interactions.

Physically, the antisymmetric vertex $(\mathbf{u}_Q \cdot K_R)\mathbf{u}_R - (\mathbf{u}_R \cdot K_Q)\mathbf{u}_Q$ is the "vortex stretching" mechanism: it rotates and amplifies the velocity field. The one-sided vertex $(\mathbf{u}_Q \cdot K_R)\mathbf{u}_R$ is pure advection without rotation. The factor-of-four difference quantifies how much vortex-stretching amplifies the perturbative coefficients relative to passive advection.

7.3 Role of scale separation

The inertial-range momenta (spanning a decade in $|\mathbf{k}|$) grow at roughly the same rate as the Kolmogorov shell (~ 0.95 vs ~ 1.2 decades/step), but with a much lower starting value (the $n = 100$ current is ~ 30 decades smaller). This is because the large-scale propagators $s_P \sim k_{\max}^2 \sim 100$ suppress the early terms in the recursion, delaying the onset of strong growth.

The Kolmogorov cascade configuration (momenta spanning two decades) tracks the Kolmogorov shell almost exactly in growth rate (~ 0.97), confirming that the asymptotic growth rate is a property of the vertex structure, not of the momentum distribution.

7.4 Lattice modes and near-resonances

The integer-momentum configuration shows dramatic spikes in the current at specific values of n (e.g., the jump from $\log_{10}|J|^2 \approx 5$ at $n = 14$ to ≈ 9 at $n = 15$, and then to ≈ 14 at $n = 16$). These occur when a partial sum K_Q of integer momenta happens to have $|K_Q|^2$ close to $\sum k_p^2$, making $s_Q \approx 0$ and producing a near-pole enhancement. This is the momentum-space signature of *near-resonances* in the Fourier decomposition of the nonlinear dynamics.

The overall growth rate for lattice modes (-0.24 decades/step over $n = 80$–100) is negative, meaning the current actually *decreases* at very high n. This is an artifact of the specific integer momentum configuration: by $n = 80$–100, the total momentum K_P is large enough that s_P grows quadratically, suppressing further growth.

7.5 Transverse dominance

In all non-abelian transverse-polarized runs, the decomposition $\mathbf{u}_P = u_\parallel \hat{K}_P + u_\perp \hat{K}_P^\perp$ shows $u_\parallel \approx 0$ to machine precision at low n (typically $|u_\parallel/u_\perp| < 10^{-14}$), confirming that the antisymmetric vertex preserves the transverse condition at each step of the recursion. At high n, numerical errors accumulate and the ratio degrades to $\sim 10^{-2}$.

This preservation of transversality by the non-abelian vertex is the Burgers analogue of incompressibility being preserved by the NS vertex. It is a consequence of the antisymmetry: the Lie bracket of two divergence-free vector fields is divergence-free.

8 Connection to the Poisson associahedra

The ordered non-abelian BG recursion (3) has the same combinatorial structure as the NNSE recursion in the main note—a sum over planar deconcatenations with a cubic vertex. The difference is:

- **NNSE vertex**: $(K_Q \times K_R)\, J_Q\, J_R$ (scalar, after transverse projection).
- **Non-abelian 2d Burgers vertex**: $(\mathbf{u}_Q \cdot K_R)\, \mathbf{u}_R - (\mathbf{u}_R \cdot K_Q)\, \mathbf{u}_Q$ (vector, no projection).

In the NNSE case, the current is scalar and the Poisson associahedron organizes the canonical form. In 2d Burgers, the current is a vector, and the analogue of the Poisson associahedron would be a *vector-valued* canonical form whose residues on the facets of the associahedron are products of vector vertex factors.

The growth-rate data suggests that the vector nature does not fundamentally change the combinatorial structure: the number of terms at each multiplicity is the same (the Catalan number C_{n-1}), and the growth rate is determined by the magnitude of the vertex factor. What changes is

that the vertex factor is now a 2×2 matrix (contracted with the vector currents), rather than a scalar (the cross product).

9 Performance

At $n = 100$, the computation involves:

- $n(n+1)/2 = 5050$ contiguous subwords,
- $\sum_{\ell=2}^{n}(n-\ell+1)(\ell-1) = n(n-1)(n+1)/6 \approx 166{,}650$ split evaluations,
- Each split requires 4 multiplications and 4 additions (two dot products and two vector updates).

The total is $\sim 1.3 \times 10^6$ floating-point operations, which runs in ~ 0.4 seconds in Python with NumPy. The memory footprint is the $(n \times n \times 2)$ current array plus the $(n \times n \times 2)$ momentum array, totaling ~ 320 KB at $n = 100$.

For comparison, the *unordered* BG recursion requires 2^n subsets and 3^n split evaluations. At $n = 100$, this is $\sim 5 \times 10^{47}$—completely infeasible. The ordered recursion sacrifices generality (it computes only one color ordering) but gains an exponential speedup.

10 Summary

1. The ordered BG recursion for 2d Burgers scales as $O(n^3)$ and reaches $n = 100$ in under half a second.
2. The transverse (vortical) sector exhibits geometric growth of the n-point current, with $|J_n|^2 \sim 10^n$ for the non-abelian vertex. This implies a finite radius of convergence for the perturbation series in the initial-data amplitude.
3. The non-abelian (antisymmetric) vertex amplifies the current $\sim 4\times$ faster than the abelian (asymmetric) vertex, quantifying the role of vortex-stretching-like dynamics in the perturbative growth.
4. The longitudinal (potential-flow) sector remains bounded due to Cole–Hopf integrability, confirming the three-tier structure: integrable $\to$ bounded; non-integrable abelian $\to$ slow growth; non-integrable non-abelian $\to$ fast growth.
5. The antisymmetric vertex preserves transversality to machine precision, mirroring how the NS vertex preserves incompressibility.
6. The growth rate is a property of the vertex structure, not of the momentum distribution: Kolmogorov shell, inertial range, and cascade configurations all give ~ 1 decade/step for the non-abelian transverse sector.